

Kartikeya Chitranshi

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PROFILE

Machine learning researcher working at the intersection of trustworthy and secure AI, adversarial robustness, and multimodal foundation models. My research focuses on worst-case and security-oriented evaluation of vision-language models, sparse and minimum-norm adversarial attacks, counterfactual learning, spurious-correlation mitigation, robust generalization, and interpretable analysis of model failures. I have two years of research experience at the Zuse Institute Berlin and an MSc in Scientific Computing from TU Berlin, with a thesis on the robustness of multimodal foundation models. My work includes a publication demonstrating that counterfactual fine-tuning of CLIP can mitigate spurious correlations more effectively than adversarial training, as well as extensive experience developing and evaluating robust machine-learning systems using PyTorch, Hugging Face Transformers, and multi-GPU infrastructure.

RESEARCH INTERESTS

Trustworthy and Secure Machine Learning; Adversarial Robustness and Machine Learning Security; Safety, Reliability, and Worst-Case Evaluation of Multimodal and Foundation Models; Sparse and Minimum-Norm Adversarial Attacks; Counterfactual Learning and Spurious-Correlation Mitigation; Robust Generalization; Explainable and Interpretable AI; and Optimization-Based Methods for Robust Learning.

PUBLICATIONS

- **Training on Plausible Counterfactuals Removes Spurious Correlations** Sadiku, S., Chitranshi, K., Kera, H., and Pokutta, S. *arXiv preprint, arXiv:2505.16583* (2025) Zuse Institute Berlin TU Berlin.[3pt] Demonstrated that counterfactual fine-tuning of CLIP removes spurious correlations more effectively than adversarial perturbations, improving worst-group accuracy by 12% over standard training on WaterBirds. Contributions included implementing counterfactual and adversarial training pipelines, fine-tuning CLIP on multi-GPU infrastructure, and evaluating robustness across image classification and image-text retrieval benchmarks.

EDUCATION

Master: MSc. in Scientific Computing
TU Berlin – Berlin, Germany

[Apr 2022] – [Jul 2025]

Final grade: 2.0 (gut)

Supervisor: Prof. Dr. Sebastian Pokutta (TU Berlin / ZIB)

Relevant coursework: High-Dimensional Optimization and Learning, Numerical Linear Algebra, Deep Learning, Machine Intelligence, Optimization under Uncertainty.

Master Thesis: [Robustness of Multi-Modal Foundation Models](#)
TU Berlin – Berlin, Germany

[Oct 2024] – [Apr 2025]

- Investigated how sparse adversarial attacks and plausible counterfactual perturbations affect vision-language models on image captioning and visual question answering tasks, finding that imperceptible perturbations can severely degrade model outputs.
- Showed that CLIP models fine-tuned on counterfactual images achieve stronger robustness than those fine-tuned on adversarially-generated images across image classification and image-text retrieval benchmarks.

- Implemented sparse adversarial attack algorithms and counterfactual training pipelines in PyTorch and HuggingFace Transformers on multi-GPU infrastructure with SLURM, enabling systematic comparison of training paradigms at scale.
- Deployed findings as an interactive public system (robust-mmfm.com) to make robustness evaluation accessible to the broader research community.

Bachelor: BSc. in Physical Sciences
University of Delhi – Delhi, India

[Jul 2018] – [Jun 2021]

WORK EXPERIENCE

Student Research Assistant

[Sep 2023] – [Sep 2025]

Zuse Institute Berlin – Berlin, Germany

Group: Prof. Sebastian Pokutta, AI in Society and Science

- Conducted research on spurious correlation robustness in vision-language models, contributing to a publication on counterfactual training (Sadiku et al., arXiv:2505.16583) as second author. Work included systematic comparison of counterfactual versus adversarial fine-tuning paradigms on CLIP models across image classification and image-text retrieval benchmarks.
- Designed and implemented counterfactual and adversarial fine-tuning pipelines for CLIP in PyTorch and HuggingFace Transformers, supporting experiments across image classification and image-text retrieval benchmarks.
- Ran large-scale distributed experiments on multi-GPU SLURM clusters, including CLIP fine-tuning, robustness evaluation, and ablation studies across training paradigms.
- Implemented sparse adversarial attack algorithms for CNNs and vision-language models, analyzing model vulnerability to small image perturbations in image captioning and visual question answering tasks.

Machine Learning Intern

[Jun 2021] – [Aug 2021]

TestAIng.com – Bangalore, India

- Implemented adversarial attack algorithms (FGSM, PGD, C&W, DeepFool) on deep neural networks in TensorFlow — first exposure to model robustness evaluation.

Data Analyst Intern

[May 2020] – [Jul 2020]

Peacock Solar – Gurugram, India

- Built automated hyperparameter optimization pipelines using Optuna for renewable energy demand forecasting.

SKILLS

- **Research Methods:** Adversarial attacks (FGSM, PGD, CW, DeepFool, sparse attacks), adversarial robustness evaluation, counterfactual training, spurious correlation analysis, supervised fine-tuning, image-text retrieval evaluation.
- **Deep Learning:** PyTorch, HuggingFace Transformers, PyTorch Lightning, TensorFlow.
- **Architectures:** Vision-Language Models (CLIP), Transformers, CNNs, Large Language Models.
- **Programming:** Python, Rust.
- **HPC:** SLURM, multi-GPU distributed training.
- **Languages:** English (C2), Hindi (native), German (A2).

ACADEMIC RECOGNITION

- Received MSc admission offers from Imperial College London, University of Edinburgh, and Trinity College Dublin (2022); chose TU Berlin.

Kantiloya Chitambar